RIPHAH INTERNATIONAL UNIVERSITY

I 14 Campus

ASSIGHMENT :

Submitted To : **Sir Imran Khan**
Subject : OOPS
Class : **CS 2:1**

Syed Husnain Shah
SAP 73745

Question no 1

import java.util.Scanner;

```
/***
 Author Husnain Shah
 SAP 73745
***/

public class PrimeNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();
        boolean isPrime = true;

        if(num <= 1) {
            isPrime = false;
        } else {
            for(int i = 2; i <= num/2; i++) {
                if(num % i == 0) {
                    isPrime = false;
                    break;
                }
            }
        }

        if(isPrime)
            System.out.println("Prime Number");
        else
            System.out.println("Not a Prime Number");
    }
}
```

Question no 2

package javaapplication3;

import java.util.Scanner;

/**

 *SAP 73745

 * @author Husnain Shah

 */

public class JavaApplication3 {

```
    public static void main(String[] args) {
        Scanner s= new Scanner(System.in);
        System.out.println("Enter Number for factorial: ");
        JavaApplication3 f = new JavaApplication3();
        int a = s.nextInt();
        System.out.println("The Factorial of " + a + " is " + f.factorial(a));

    }

    int factorial(int a){
        int fact=1;
        for(int i = 1; i < a;  i++){
            fact =  fact*i;
        }
        return fact;
    }
}
```

Question no 3

```
package arraycount;

/**
 *SAP 73745
 * @author Husnain Shah
 */
```

```
public class Arraycount
{
    public static void main(String[] args)
    {

        int arr[] = {10, 20, 30, 40, 50};

        int length = arr.length;

        System.out.println("Length of array = " + length);
    }

}
```

Question No 4

```
package palyndrom;
import java.util.Scanner;
/**
 *SAP 73745
 * @author Husnain Shah
 */
public class Palyndrom
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        int original = num;
        int reverse = 0;

        while(num > 0)
        {
            int digit = num % 10;
            reverse = reverse * 10 + digit;
            num = num / 10;
        }

        if(original == reverse)
            System.out.println("Palindrome Number");
        else
            System.out.println("Not a Palindrome Number");
    }
}
```

Question No 5

```
package swapnumbers;
import java.util.Scanner;

/**
 *SAP 73745
```

```
 * @author Husnain Shah
 */
public class Swapnumbers {

    /**
     * @param args the command line arguments
     */
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        a = a + b;
        b = a - b;
        a = a - b;

        System.out.println("After swapping:");
        System.out.println("a = " + a);
        System.out.println("b = " + b);
    }
}
```

Question no 6

```
package starpattern;


/**
 *SAP 73745
 * @author Husnain Shah
 */
public class Starpattern {


    public static void main(String[] args) {
        for(int i = 1; i <= 5; i++) {
            for(int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }


}
```